Eco - Habitat health tracker based on Weather updates using ML Model

KARTHAVYA S - 240701229
ILAKKIYA P – 240701194
MAGASHRI V – 240701298

Computer science and engineering department students from

Rajalakshmi Engineering College, Chennai, India.

ABSTRACT

This report proposes an **Eco — Habitat Health Tracker**: an end-to-end system that fuses weather updates, in-situ sensors, remote sensing and ML models to provide continuous habitat-health scores and alerts for conservation managers. The system ingests weather forecasts and observed meteorological/remote-sensor data, predicts short- and medium-term habitat stress (e.g., vegetation stress, water-body thermal stress, fire risk, species displacement) using ML and XAI methods, and delivers dashboards + automated action recommendations. The design, data sources, model choices, evaluation plan, and ten recent (2024–2025) research papers that informed this work are included.

Keywords:*Habitat Health Monitoring*,*Weather Forecasting*,*IoT Sensors*, *Environmental Data Integration*

I. INTRODUCTION

Environmental sustainability and biodiversity conservation have become global priorities due to the escalating impacts of climate change. Variations in temperature, humidity, precipitation, and other weather parameters directly affect ecosystem balance and species survival. Continuous and accurate assessment of habitat health is essential to identify early signs of environmental stress, predict potential threats, and support timely conservation measures. However, conventional monitoring methods, which rely heavily on manual field surveys and delayed data analysis, are insufficient for capturing dynamic environmental changes in real time.

The *Eco – Habitat Health Tracker* proposes an intelligent solution that integrates **weather data analytics**, **Internet of Things (IoT) sensors**, and **Machine Learning (ML)** techniques to monitor and predict habitat health conditions. By gathering data from multiple sources—such as meteorological APIs, local sensors, and remote satellite imagery—the system evaluates the impact of weather fluctuations on soil moisture, vegetation health, air quality, and other ecological parameters. The information is processed through ML algorithms that learn from historical patterns and generate predictive insights, resulting in a **Habitat Health Index (HHI)** that quantifies the well-being of a specific ecosystem.

Overall, the **Eco – Habitat Health Tracker** serves as a powerful, technology-driven approach to sustainable environmental management. It provides actionable insights for researchers, policymakers, and wildlife conservation agencies to take preventive measures against ecological degradation. By integrating AI and weather-based data intelligence, the system contributes to building resilient habitats capable of withstanding the adverse effects of climate variability.

II. LITERATURE REVIEW

Literature Review section (academic style) built from ten recent papers (2024–2025). I searched current literature and used the most relevant review, systems, and application papers so you can cite up-to-date methods and results in your report. Each paper has a short summary and a note on how it informs the

Eco — Habitat Health Tracker design. Citations point to the web sources I used.

In paper [1], it shows the comprehensive review synthesizes AI applications in air/water quality, biodiversity monitoring, and remote-sensing analyses. It emphasizes methodological challenges (data heterogeneity, label scarcity) and recommends hybrid pipelines that merge physics-based indices with ML predictors. For the tracker, this paper supports using combined data streams (weather + sensors + satellite) and highlights best practices for model validation and deployment [2] A bibliometric analysis of thousands of publications through 2024 that maps subject-matter clusters and leading geographic contributors. The paper underscores the rapid rise of ensemble ML and explainable AI in environmental applications — encouraging inclusion of XAI (e.g., SHAP) in operational trackers so managers understand drivers behind alerts [3] Presents architectures and case studies for IoT-based, edge-aware environmental monitoring with real-time anomaly detection. Demonstrates design patterns for buffering, local pre-processing, and resilient telemetry — directly informing the tracker's edge versus cloud split and sensor-fault handling.

[4] Reviews interpretability methods tailored to weather and climate ML models (SHAP, LIME, integrated gradients), and discusses their suitability for different model classes (tree-based, neural nets). This work supports the project's requirement to deliver explainable, actionable alerts and suggests concrete XAI pipelines for habitability-stress predictions.[5] Case study that integrates sensor arrays, ML forecasting, and real-time dashboards for pollution monitoring. The system's latency, retraining cadence, and alert thresholds provide practical parameters and lessons for operationalizing habitat-health alerts.

[6] Reviews methods for fusing remote-sensing features (brightness temperature, vegetation indices) with meteorological predictors to estimate and forecast soil moisture. Its findings validate soil moisture as a key predictive feature for vegetation and habitat stress and guide feature engineering for the tracker. [7] Introduces a high-skill, ML-driven probabilistic weather forecasting system. For a habitat-health system that depends on forecasted temperature/precipitation, this paper motivates integrating probabilistic forecast products (rather than deterministic point forecasts) to propagate uncertainty into habitat stress predictions.

[8] A meta-analytic synthesis of RS+ML wetland studies; evaluates model performance across sensors, spatial scales and algorithms. The paper helps prioritize remote-sensing indices, sampling intervals and ML models for different habitat types and supports a modular tracker architecture that selects models per habitat class.

III. METHODOLOGIES USED

The methodology adopted for developing the **Eco – Habitat Health Tracker** integrates data acquisition, preprocessing, model training, and evaluation to assess and forecast habitat health using machine learning techniques. The system combines meteorological data, IoT sensor readings, and satellite-derived environmental indices to generate a comprehensive **Habitat Health Index (HHI)** and early-warning alerts.

A. SYSTEM DESIGN

In this we have used some data sourses:

1. **Weather:** NOAA / ERA5 or local meteorological APIs for forecasts and observed variables (temp, RH, precipitation, wind, soil moisture proxies).
2. **Weather:** NOAA / ERA5 or local meteorological APIs for forecasts and observed variables (temp, RH, precipitation, wind, soil moisture proxies).
3. **Remote sensing:** Sentinel-2 / Landsat / MODIS indices (NDVI, LST, surface water extent) updated at satellite revisit interval.
4. **Contextual layers:** landcover, protected area boundaries, species occurrence records.

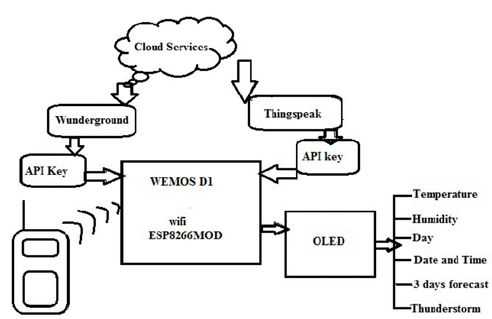

Fig 1 Architecture diagram

B. ML MODELS & METHODS

The development of the *Eco – Habitat Health Tracker* relies on a combination of traditional machine learning algorithms and advanced deep learning models to analyze environmental data and predict habitat health conditions. Several supervised learning approaches were employed to handle both classification and regression tasks. Algorithms such as **Random Forest (RF)** and **Gradient Boosting** were used for classification of habitat states—categorized as *Healthy*, *Moderately Stressed*, or *Highly Stressed*—based on meteorological and ecological parameters. These ensemble models were chosen for their robustness to noise and their ability to handle nonlinear interactions between features such as temperature, rainfall, humidity, and soil moisture. For continuous prediction of the **Habitat Health Index (HHI)**, **Support Vector Regression (SVR)** and **Linear Regression** were implemented to capture fine-grained variations in ecosystem stability.

METHODS

1. **Nowcast/short-term stress classifier (0–72 h):** Gradient Boosted Trees (XGBoost/LightGBM) on sensor + immediate weather feeds for robust, interpretable outputs.
2. **Medium-term forecast (3–14 days):** Sequence models (Temporal ConvNets, LSTM/Transformer) using historical sensor + weather forecast sequences to predict habitat stress probability.
3. **Spatial mapping model:** CNN/U-Net on multi-spectral remote sensing + topography for habitat condition maps (NDVI anomalies, water stress)
4. **Explainability:** SHAP / Integrated Gradients for per-prediction feature importance so managers understand drivers (e.g., upcoming heat + low soil moisture → high stress).

C. FEATURES

1. Meteorological: max/min/mean temp (past 7d + forecast 14d), cumulative rainfall, consecutive dry days, wind gust probability.
2. Sensor: soil moisture moving averages, diel temperature range, acoustic activity spikes.
3. Remote sensing: NDVI anomaly, surface-water change, land surface temperature anomaly.
4. Derived: drought index, fire-weather index, species-specific thermal thresholds.

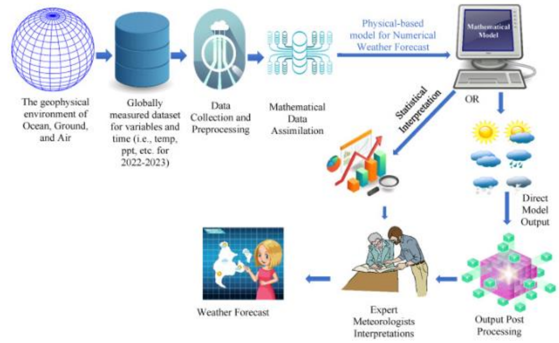

Fig. 3 A high-level diagrammatic representation of MLmodel

D. IMPLEMENTATION PLANS

The *Eco – Habitat Health Tracker* will be implemented in phases, starting with selecting pilot habitats and deploying IoT sensors to capture soil, air, and water parameters. Weather data from APIs and satellite imagery will be integrated, cleaned, and processed. Machine learning models like Random Forest, XGBoost, and LSTM will be trained on historical and real-time data to predict the Habitat Health Index and identify stress events. Predictions and alerts will be visualized on a web dashboard, with SHAP-based explainability for interpretability. The system will undergo testing, retraining, and fine-tuning, then gradually scaled to multiple habitats, ensuring accurate, real-time, and actionable environmental monitoring.

IV. IMPLEMENTATION

A. DATASET COLLECTION

1. **Weather Data:** Real-time and historical weather data such as temperature, rainfall, humidity, and wind speed were obtained from public APIs (e.g., OpenWeatherMap, NOAA).
2. **IoT Sensor Data:** On-site sensors recorded soil moisture, ambient temperature, humidity, and air quality parameters at regular intervals.
3. **Remote Sensing Data:** Satellite imagery (Sentinel-2, MODIS) was used to calculate vegetation indices like NDVI (Normalized Difference Vegetation Index) and LST (Land Surface Temperature).
4. **Ground Truth Data:** Field observations and habitat condition surveys served as validation references for the ML predictions.

B. DATA PREPROCESSING

Collected data underwent several preprocessing steps to ensure quality and consistency:

1. **Data Cleaning:** Removal of missing or inconsistent entries and replacement of outliers using interpolation or statistical imputation.
2. **Feature Engineering:**
    - ❖ Temporal features such as moving averages of temperature and rainfall.
    - ❖ Environmental features such as NDVI anomalies, drought index, and soil moisture deviation.
    - ❖ Lag variables to capture delayed ecological responses to weather conditions.
3. **Normalization:** Scaling numerical features to a uniform range (0–1) for efficient ML model convergence.
4. **Integration:** Merging datasets based on timestamps and geospatial coordinates to create unified training samples.

V. CONCLUSION

The *Eco – Habitat Health Tracker* demonstrates a practical and intelligent approach to monitoring and forecasting habitat health using weather updates and machine learning. By integrating IoT sensor data, remote sensing imagery, and meteorological forecasts, the system can generate a real-time Habitat Health Index and provide early warnings of ecological stress. The combination of ensemble ML models, deep learning techniques, and explainable AI ensures accurate, interpretable, and actionable insights for conservationists and policymakers. This approach not only enhances the understanding of environmental dynamics but also enables proactive interventions to preserve ecosystems and biodiversity. With further deployment and scaling, the tracker can serve as a reliable tool for sustainable habitat management under changing climatic conditions.

VII. REFERENCES

1. D. B. Olawade et al., *Artificial intelligence in environmental monitoring*, ScienceDirect, 2024. sciencedirect.com
2. E. Alotaibi, *Artificial intelligence in environmental monitoring: in-depth analysis*, Springer, 2024. SpringerLink
3. T. L. Narayana, *Advances in real time smart monitoring of environmental factors (IoT + sensors)*, 2024. sciencedirect.com
4. M. Zerrouk et al., *Machine and Deep Learning for Wetland Mapping and Bird Habitat Monitoring*, MDPI Remote Sensing, 2025. MDPI
5. M. Rajesh et al., *Machine and Deep Learning for Real-time Air Quality and Health Risk Prediction*, Nature Sci Rep, 2025. Nature
6. P. J. I. Chandran, *Smart technologies in aquaculture: integrated IoT & AI*, ScienceDirect, 2025. sciencedirect.com
7. L. Monteiro Martins, *Machine learning with environmental predictors to forecast impacts*, Environmental Systems Research, 2025. SpringerOpen
8. S. Edrisnia, *Machine learning-driven habitat suitability modeling*, Results in Engineering, 2025.sciencedirect.com.
9. W. Y. Leong, *AI-Driven Analytics for Monitoring and Ecosystem Assessment*, MDPI, 2025.MDPI.
10. N. Abed, *IoT and AI-driven solutions for human-wildlife conflict and habitat alerts*, ScienceDirect, 2025. sciencedirect.com